

MYSQL FUNDAMENTALS

INSTALLATION, DATABASES, TABLES, DATA TYPES, CONSTRAINTS

1. INTRODUCTION TO DATABASES

Before writing SQL, one must understand what problem SQL solves.

A database is an organized collection of data stored in a structured format so that it can be easily accessed, managed, and updated.

Key Terms

- **Database** → Container that holds tables
- **Table** → Structure that stores data in rows and columns
- **Row (Record)** → One complete entry
- **Column (Field)** → One attribute of the data

Real-world Analogy

Concept	Real-world
Database	School system

Table	Student list
Row	One student
Column	Name, Age, Marks

2. INSTALLATION OF MYSQL

We will use:

- **MySQL** (Database Engine)
- **MySQL Workbench** (Interface)

Steps (Conceptual Flow)

1. Install MySQL Server
2. Set root password
3. Install Workbench
4. Connect to local instance

3. SQL BASICS – WHAT IS SQL?

SQL = Structured Query Language

Used for:

- Creating databases
- Creating tables
- Inserting data
- Querying data
- Updating data
- Deleting data

4. DATABASE OPERATIONS

4.1 CREATE DATABASE

CREATE DATABASE company;

Explanation

- **CREATE** → Command used to create objects

- **DATABASE** → Specifies we are creating a database
- **company** → Name of the database

4.2 SHOW DATABASES

SHOW DATABASES;

Explanation

- Displays all databases present in the MySQL server

4.3 USE DATABASE

USE company;

Explanation

- Selects the database to work on
- Without this, MySQL doesn't know where to create tables

4.4 DROP DATABASE

DROP DATABASE test_db;

Explanation

- Deletes the database permanently

- Use carefully (no undo)
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5. TABLE CREATION

A table defines structure before storing data.

5.1 BASIC TABLE CREATION

```
CREATE TABLE employees ( id INT, name VARCHAR(50), salary FLOAT, joining_date DATE );
```

Explanation

- **CREATE TABLE** → Command to create table
- **employees** → Table name

Inside brackets:

- **id INT**
 - Column name: id
 - Data type: INT (integer values)
- **name VARCHAR(50)**
 - Variable-length string (max 50 characters)
- **salary FLOAT**

- Stores decimal numbers
 - **joining_date DATE**
 - Stores date (YYYY-MM-DD)
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6. DATA TYPES (VERY IMPORTANT)

Numeric Types

- **INT** → Whole numbers
- **FLOAT** → Decimal numbers

String Types

- **VARCHAR(n)** → Variable-length text
- **CHAR(n)** → Fixed-length text

Date Types

- **DATE** → YYYY-MM-DD
 - **DATETIME** → Date + time
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7. NULL CONCEPT

- **NULL** means "no value"
- It is **NOT** equal to 0 or empty string

Example:

```
INSERT INTO employees (id, name) VALUES (1, NULL);
```

8. CONSTRAINTS

Constraints enforce rules on data.

8.1 PRIMARY KEY

```
CREATE TABLE employees ( id INT PRIMARY KEY, name VARCHAR(50) );
```

Explanation

- **PRIMARY KEY:**
 - Uniquely identifies each row
 - Cannot be NULL
 - Cannot have duplicate values

8.2 NOT NULL

name VARCHAR(50) NOT NULL

Explanation

- Ensures column must have a value

8.3 UNIQUE

email VARCHAR(100) UNIQUE

Explanation

- Prevents duplicate values

8.4 DEFAULT

salary FLOAT DEFAULT 30000

Explanation

- Assigns default value if none provided

9. FOREIGN KEY (INTRODUCTION)

Used to connect tables.

Example

```
CREATE TABLE departments ( dept_id INT PRIMARY KEY, dept_name VARCHAR(50) );
```

```
CREATE TABLE employees ( id INT PRIMARY KEY, name VARCHAR(50), dept_id INT, FOREIGN KEY (dept_id)  
REFERENCES departments(dept_id) );
```

Explanation

- **dept_id** in employees references departments table
- Ensures:
 - No invalid department values
 - Maintains data integrity

10. DROP TABLE

```
DROP TABLE employees;
```

Explanation

- Deletes table permanently

11. FULL PRACTICE FLOW

SQL

-- Create database

```
CREATE DATABASE company;
```

-- Select database

```
USE company;
```

-- Create departments table

```
CREATE TABLE departments (
```

```
    dept_id INT PRIMARY KEY, -- Unique department ID
```

```
    dept_name VARCHAR(50) NOT NULL -- Department name cannot be NULL
```

);

-- Create employees table

CREATE TABLE employees (

id INT PRIMARY KEY, -- Unique employee ID

name VARCHAR(50) NOT NULL, -- Employee name

salary FLOAT DEFAULT 30000, -- Default salary

dept_id INT, -- Foreign key column

FOREIGN KEY (dept_id) REFERENCES departments(dept_id) -- Relationship

);

12. COMMON MISTAKES (VERY IMPORTANT)

1. Forgetting **USE** database
 2. Missing comma between columns
 3. Wrong data type selection
 4. Not defining primary key
 5. Misunderstanding NULL vs empty string
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13. MINI ASSIGNMENT

1. Create database **school**
 2. Create table **students** with:
 - id (Primary Key)
 - name (Not Null)
 - age
 - marks (default 0)
 3. Create table **courses**
 4. Add foreign key in students referencing courses
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14. EXPECTED UNDERSTANDING AFTER DAY 1

Students should:

- Understand database vs table
 - Create and delete databases
 - Create tables with proper structure
 - Use data types correctly
 - Apply constraints
 - Understand relationships (basic level)
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